

Algorithm complexity from the iterative loop in a code

Total points 13/15 ?

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0 of 0 points

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Quiz

13 of 15 points



✓ Analyze the following code. Note the relationship between i and j. *

1/1

```
int j = 0;
for (int i = 0; i < n; i++) {
    while (j < n && A[i] < B[j]) {
        j++;
    }
}
```

- ☒ $O(n)$
- ☐ $O(1)$
- ☐ $O(n^2)$
- ☐ $O(n \log n)$



✓ Analyze the code and provide the tightest upper bound: *

1/1

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j = j + i) {
        for (int k = 1; k <= 100; k++) {
            // Constant time
        }
    }
}
```

- ☐ $O(100n^2)$
- ☒ $O(n \log n)$
- ☐ $O(n^3)$
- ☐ $O(n^2)$



✗ A loop runs from 1 to n , and inside it, another loop runs from 1 to n^2 . However, the inner loop only executes when i is a multiple of 10. What is the overall complexity? *0/1

- ☐ $O(n^2)$
- ☐ $O(n)$
- ☒ $O(n^3)$
- ☐ $O(n^2 / 10)$

✗

Correct answer

- ☒ $O(n^2)$

✓ Determine the time complexity of the following nested loop where the inner loop depends on the outer loop variable: *1/1

```
for (int i = 1; i <= n; i++) {  
    for (int j = 1; j <= i; j++) {  
        count++;  
    }  
}
```

- ☒ $O(n^2)$
- ☐ $O(n)$
- ☐ $O(n \log n)$
- ☐ $O(n!)$

✓



✓ What is the asymptotic upper bound of the following code? *

1/1

```
for (int i = 1; i <= n; i++) {  
    for (int j = 1; j <= n; j += i) {  
        // Constant time operation  
    }  
}
```

- ☐ $O(\log n)$
- ☐ $O(n^2)$
- ☒ $O(n \log n)$
- ☐ $O(n)$



✓ Evaluate the complexity of the following loop with a square root termination condition:

*1/1

```
for (int i = 0; i * i < n; i++) {  
    // Constant time operation  
}
```

- ☐ $O(n^2)$
- ☒ $O(\sqrt{n})$
- ☐ $O(n)$
- ☐ $O(\log n)$



- ✓ Consider a loop where the control variable increases exponentially. Find the Big O complexity: *1/1

```
for (int i = 2; i <= n; i = pow(i, 2)) {
    // Constant time operation
}
```

- ☐ $O(\sqrt{n})$
- ☐ $O(n)$
- ☐ $O(\log n)$
- ☒ $O(\log \log n)$



- ✗ Consider the following code where the inner loop limit grows geometrically. What is the complexity? *0/1

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= pow(2, i); j++) {
        // Constant time
    }
}
```

- ☐ $O(n^2)$
- ☒ $O(2^n)$
- ☐ $O(2^{(n+1)})$
- ☐ $O(n * 2^n)$



Correct answer

- ☒ $O(2^{(n+1)})$



- ✓ What is the worst-case time complexity of the following snippet involving a conditional jump? *1/1

```
for (int i = 0; i < n; i++) {  
    if (i % 2 == 0) {  
        for (int j = 0; j < n; j++) {  
            sum++;  
        }  
    }  
    else {  
        sum++;  
    }  
}
```

- ☐ $O(n \log n)$
- ☒ $O(n^2)$
- ☐ $O(2^n)$
- ☐ $O(n)$



✓ Calculate the total number of iterations for the following loop structure: * 1/1

```
for (int i = n; i > 0; i /= 2) {  
    for (int j = 0; j < i; j++) {  
        // Constant time operation  
    }  
}
```

☒ $2n$



☐ $\log n$

☐ $n \log n$

☐ n^2

✓ Analyze the following code snippet and determine the exact number of times the printf statement is executed. *1/1

```
for (int i = 0; i < n; i++) {  
    for (int j = 0; j < m; j++) {  
        printf("Hello");  
    }  
}
```

☐ $m^{\log n}$

☒ $n * m$



☐ $n + m$

☐ n^2



✓ Determine the complexity class of the following iterative process: *

1/1

```
int i = n;  
while (i >= 1) {  
    for (int j = 0; j < n; j++) {  
        // Constant time  
    }  
    i = i / 3;  
}
```

- ☐ $O(n)$
- ☐ $O(3^n)$
- ☐ $O(n^2)$
- ☒ $O(n \log n)$



✓ Which of the following best describes the complexity of this combination of loops? *1/1

```
for (int i = 0; i < n; i++) {  
    for (int j = n; j > i; j--) {  
        // Step 1  
    }  
}  
for (int k = 1; k < n; k = k * 2) {  
    // Step 2  
}
```

- ☒ $O(n^2 + \log n)$ which simplifies to $O(n^2)$
- ☐ $O(n^2 + n)$
- ☐ $O(n^2 * \log n)$
- ☐ $O(n \log n)$



✓ What is the time complexity of the following loop where the increment is multiplicative? *1/1

```
for (int i = 1; i <= n; i = i * 2) {  
    // Constant time operation  
}
```

- ☐ $O(2^n)$
- ☐ $O(n \log n)$
- ☐ $O(n)$
- ☒ $O(\log n)$



- ✓ Find the closed-form expression for the number of primitive operations in *1/1 this triple nested loop:

```
for (int i = 1; i <= n; i++) {  
    for (int j = 1; j <= i; j++) {  
        for (int k = 1; k <= j; k++) {  
            count++;  
        }  
    }  
}
```

- ☐ $(n^2 + n) / 2$
- ☒ $n(n+1)(n+2)/6$
- ☐ $n \log n$
- ☐ $n^3 / 2$



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